

MASTER 1 THESIS — GEOMATICS AND SPATIAL MODELLING

Dynamics Monitoring System: A Multi-Temporal Geomatics Approach to Monitoring the 25 Forest Massifs of the Bouches-du-Rhône, Southern France, 2010–2025

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Presentation Outline

1 Context and Problem Statement

2 Study Area

3 Research Objectives

4 Data and Methodology

5 Results

6 Discussion, Limitations and
Implications

7 Conclusion and Perspectives

01- Context and Problem Statement

- **Mediterranean ecosystems under growing pressure**

The Mediterranean basin is warming 20% faster than the global mean, driving structural summer droughts and severe canopy dieback.

- **Expanding Wildland-Urban Interfaces (WUI)**

Rapid urban sprawl around the Aix-Marseille conurbation densifies the contact between flammable vegetation and dwellings, amplifying ignition risk.

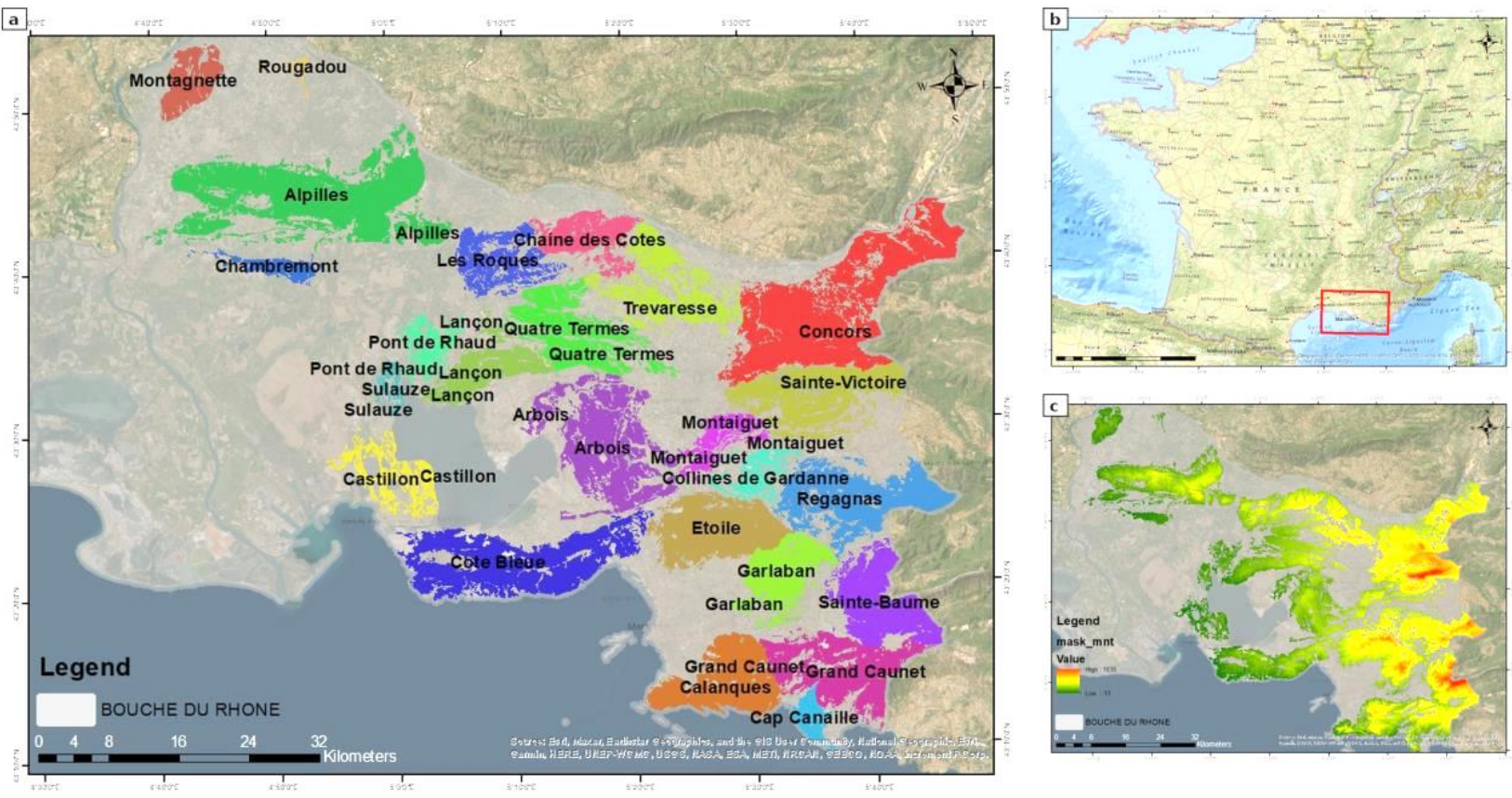
- **A need for reproducible, department-wide monitoring**

Les services forestiers et de prévention des incendies (DFCI, DDTM13) ont besoin d'un système automatisé multi-capteurs couvrant les 25 massifs classés.

Problem statement *How can optical, radar and LiDAR observation be articulated within geomatics to characterize — and distinguish — chronic ecological decline from abrupt disturbance (wildfire) across the 25 forest massifs?*

02-STUDY AREA

Figure 1. Geographic Setting of the Bouches-du-Rhone Forest Massifs



(a) The 25 officially classified DFCI forest massifs · (b) regional location · (c) SRTM-derived elevation

25

officially classified forest massifs

5,087 km²

of forested territory

Contrasting relief — alluvial plains, coastal garrigue and limestone ridges — strongly conditions forest distribution and dominant typologies (Pinus halepensis, Quercus ilex/pubescens).

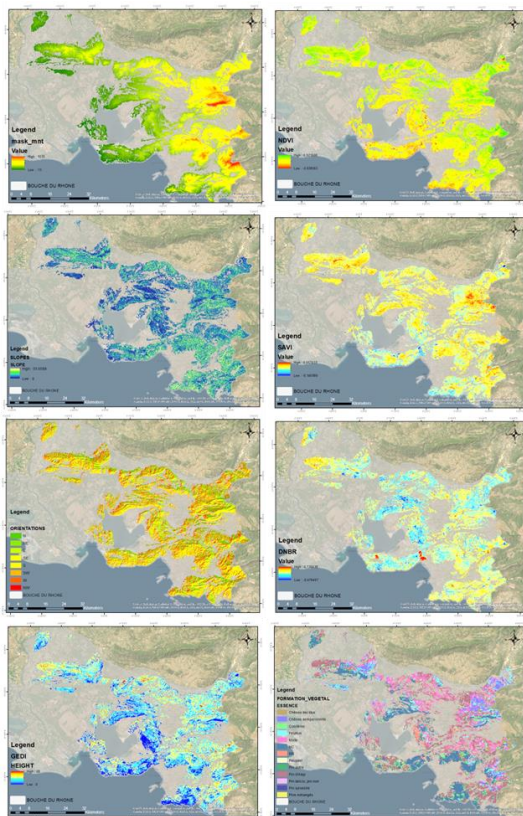
03- OBJECTIVES

Scientific Objectives

-  Build an automated, cloud-masked, and radiometrically harmonized Landsat 5/7/8/9 surface reflectance time series restricted to the May–June phenological window.
-  Calculate multi-index trajectories (NDVI, NDMI, SAVI, NBR/dNBR) and model pixel-level monotonic trends using non-parametric Mann-Kendall and Theil-Sen estimators.
-  Implement LandTrend based temporal segmentation to decouple abrupt wildfire scars from progressive drought-induced browning.
-  Integrate Sentinel-1 C-band SAR backscatter and NASA GEDI LiDAR structural profiles (RH98, AGBD) to validate physical canopy degradation.

03- OBJECTIVES

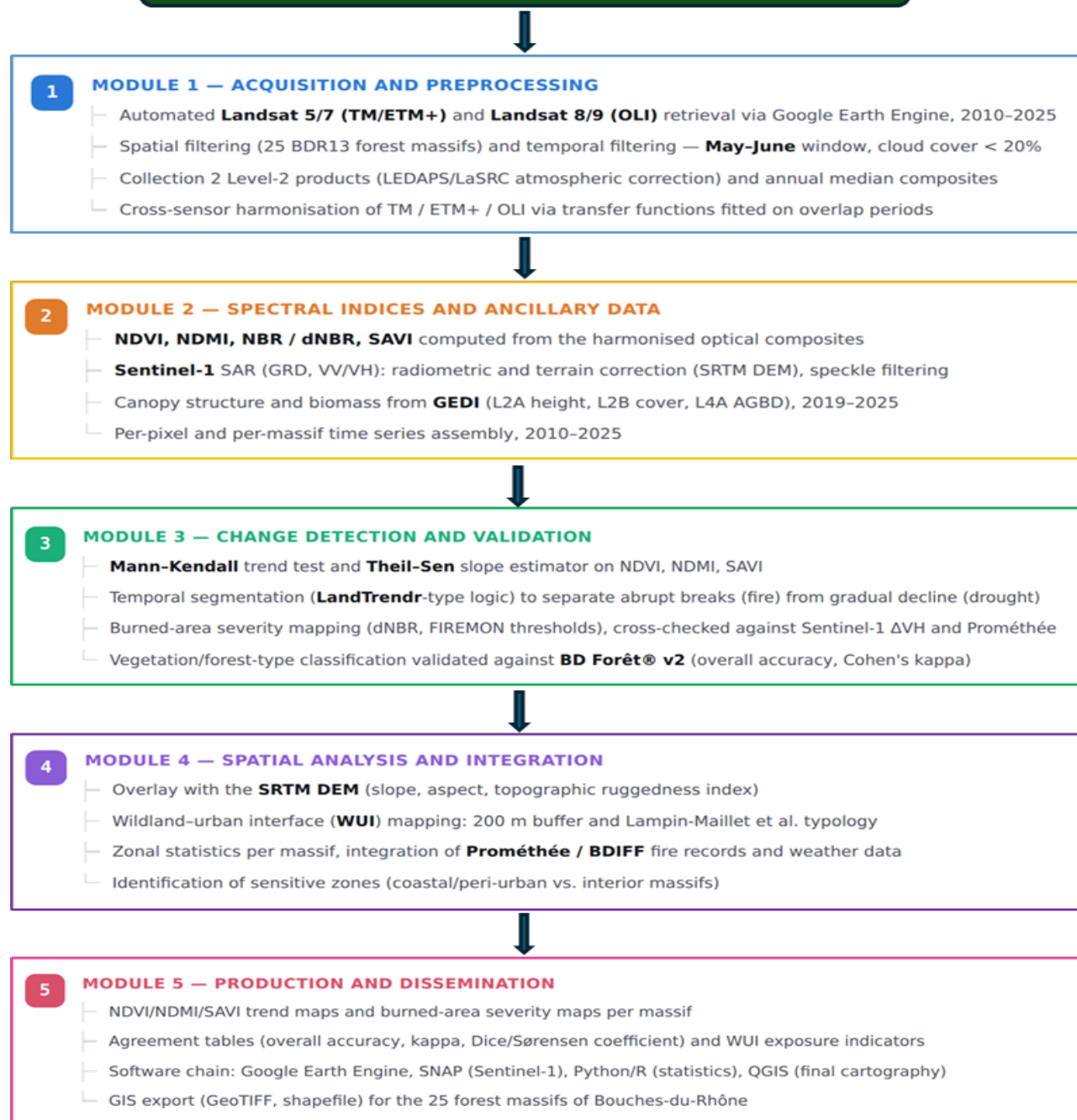
Remote Sensing and Ancillary Datasets



Platform / Sensor	Period	Spatial Res.	Revisit	Spectral Bands	Processing Level
Landsat 5 TM	2010–2011	30 m	16 days	B, G, R, NIR, SWIR1, SWIR2	Collection 2 Tier 1 L2 (LEDAPS)
Landsat 7 ETM+	2010–2025	30 m	16 days	B, G, R, NIR, SWIR1, SWIR2	Collection 2 Tier 1 L2 (LEDAPS)
Landsat 8 OLI	2013–2025	30 m	16 days	Ultra Blue, B, G, R, NIR, SWIR1, SWIR2	Collection 2 Tier 1 L2 (LaSRC)
Landsat 9 OLI-2	2021–2025	30 m	16 days	Ultra Blue, B, G, R, NIR, SWIR1, SWIR2	Collection 2 Tier 1 L2 (LaSRC)
Sentinel-1 SAR	2014–2025	10 m	6–12 days	C-band (5.405 GHz), VV + VH	Level-1 GRD ($\sigma_0 \rightarrow \gamma_0$)
NASA GEDI	2019–2025	~25 m	Profiling	1064 nm LiDAR waveforms	Level 2A, 2B, 4A (footprints)
SRTM DEM	2000	30 m	Static	InSAR Topography	1 Arc-Second Global V3
BD Forêt® v2	2018–2024	Vector	Static	National Forest Inventory polygons	Structural forest typology
Prométhée / BDIFF	2010–2025	Vector/Tab	Event	Fire perimeters, ignition dates	Official French Fire Registry

04- METHODOLOGY

Monitoring of Forest Massifs in BDR (2010–2025)



04- METHODOLOGY

Phenological Window and Cross-Sensor Harmonization

Strict May 1st – June 30th window

- Photosynthetic peak of Mediterranean vegetation following spring rainfall
- Leaf flushing of deciduous oaks (*Quercus pubescens*) is complete
- Herbaceous understory still green, ahead of summer curing
- Minimal topographic shadowing (solar zenith angles near their annual maximum)

Radiometric Harmonization (Roy et al., 2016)

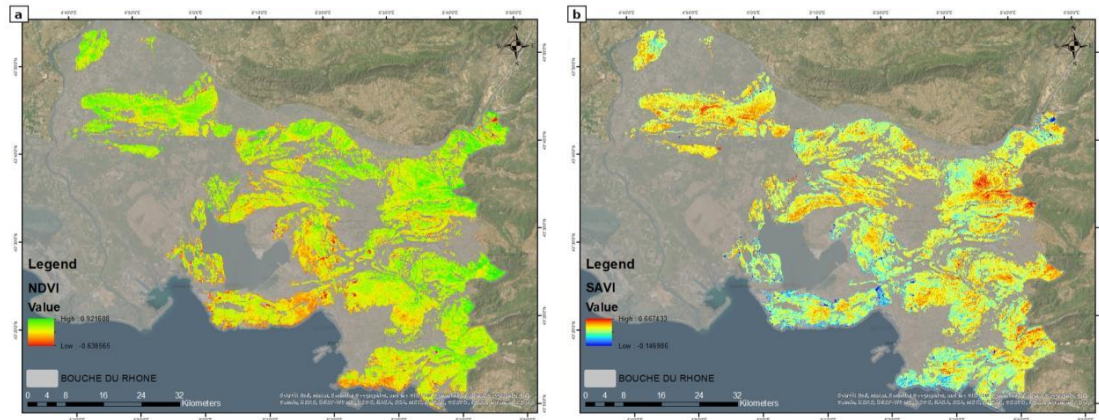
Linear transfer functions (OLI → ETM+), applied band-by-band to ensure radiometric continuity between Landsat 5/7 and 8/9:

Cloud/shadow masking via CFMask (Foga et al., 2017) · cloud cover < 20% · USGS Collection 2 surface reflectance (LEDAPS/LaSRC)

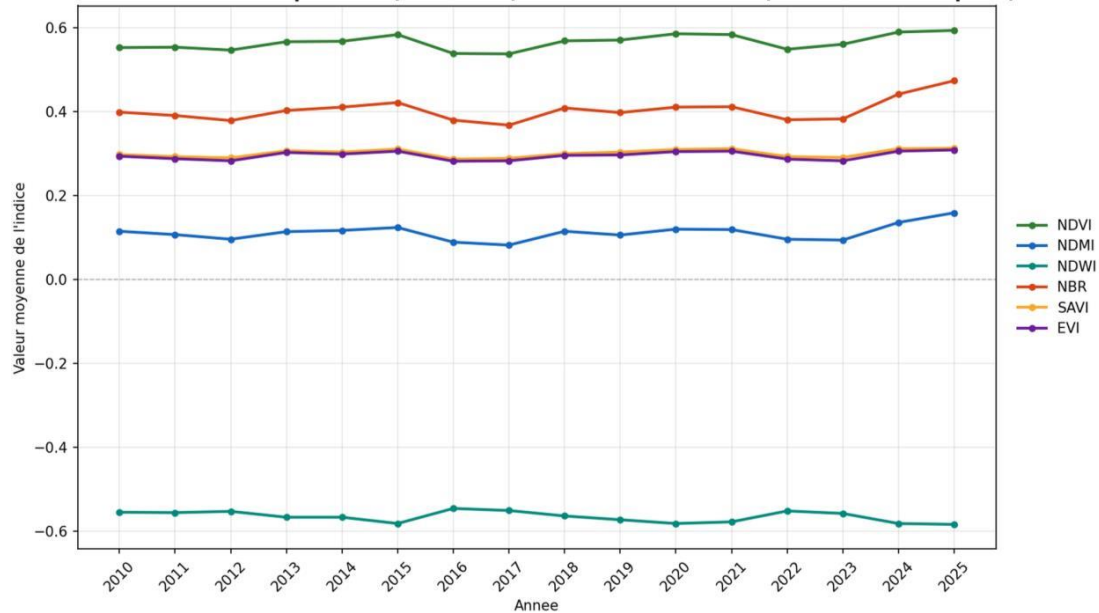
Landsat is the only sensor common to both 2010 and 2025: it forms the indispensable backbone of the multi-temporal comparison despite a resolution three times coarser than Sentinel-2

05- RESULTS

Spectral Trajectories 2010–2025



c) Evolution des indices spectraux (2010-2025) - 25 massifs forestiers (zone d'étude complète)



- NDVI (greenness/vigor) and SAVI (soil-brightness adjusted) — panels (a) and (b)
- NDMI (canopy moisture) and NBR/dNBR (burn severity), used throughout the following results
- Pixel-level trends estimated by Theil-Sen, significance tested by Mann-Kendall
- Landsat is the only sensor common to both 2010 and 2025: it forms the indispensable backbone of the multi-temporal comparison despite a resolution three times coarser than Sentinel-2

05- RESULTS

Massif-Level Trends: Resilience vs. Vulnerability

Department-wide trend modeling (2010–2025) resolves sharp massif-to-massif divergence in vegetation trajectory and disturbance exposure.

Most resilient massifs (β_{NDVI})

Sainte-Victoire	$+1.84 \times 10^{-3}/\text{yr}$	4.2% degraded
Sambuc	$+1.12 \times 10^{-3}/\text{yr}$	4.8% degraded
Concors	$+1.42 \times 10^{-3}/\text{yr}$	5.1% degraded
Petit Luberon	$+1.05 \times 10^{-3}/\text{yr}$	4.1% degraded
Sainte-Baume	$+0.95 \times 10^{-3}/\text{yr}$	3.1% degraded

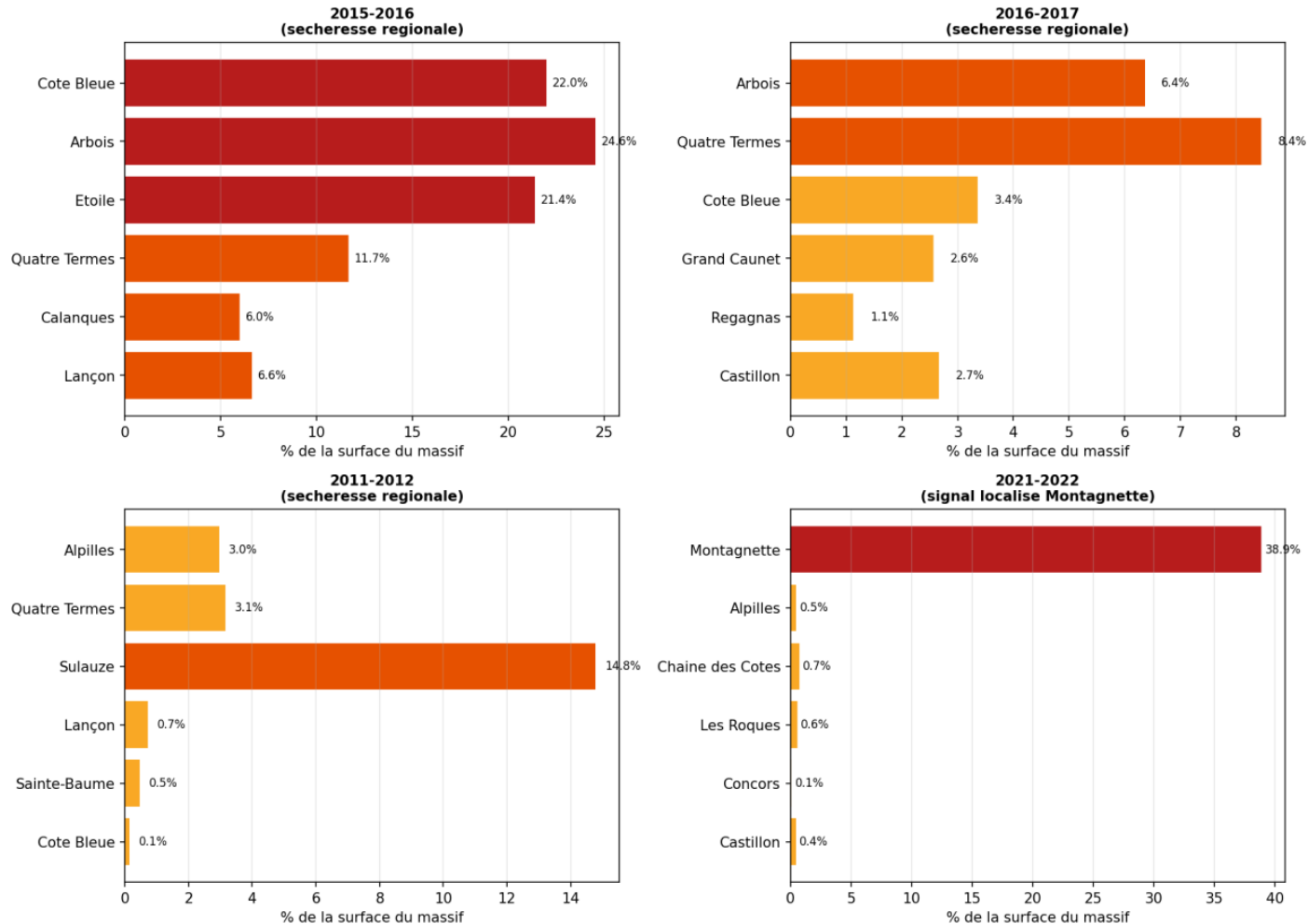
Most vulnerable massifs (β_{NDVI})

Castillon	$-2.74 \times 10^{-3}/\text{yr}$
36.5% degraded, 1,020 ha burned	
Marignane	$-2.15 \times 10^{-3}/\text{yr}$
31.2% degraded, 210 ha burned	
Calanques	$-1.92 \times 10^{-3}/\text{yr}$
28.1% degraded, 410 ha burned	
Côte Bleue	$-1.85 \times 10^{-3}/\text{yr}$
24.3% degraded, 740 ha burned	
Étoile	$-1.45 \times 10^{-3}/\text{yr}$
22.4% degraded, 2,850 ha burned	

05- RESULTS

Disturbance by Period (dNBR > 0.1)

Massifs les plus touchés par période (% de la surface du massif classée dNBR > 0.1)

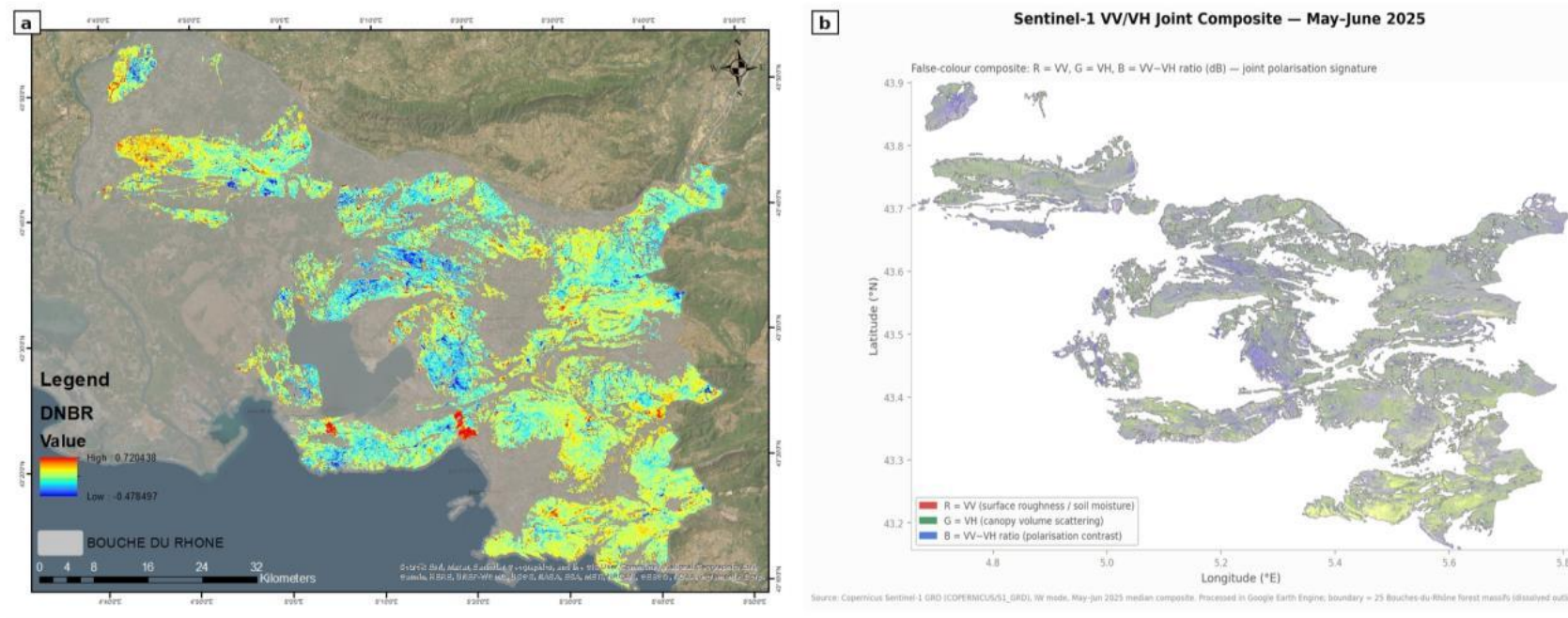


Regional drought episodes vs. a localized signal

- 2011-2012, 2015-2016, 2016-2017: regionally synchronous, drought-driven browning (Arbois, Côte Bleue, Étoile, Quatre Termes, Sulauze, among others)
- 2021-2022: a spatially localized but intense signal, concentrated at the Montagnette massif (38.9% of its surface classified dNBR > 0.1)
- Period-based screening distinguishes diffuse regional stress from a circumscribed, single-massif event

05- RESULTS

Burn Severity (dNBR) and Sentinel-1 Composite

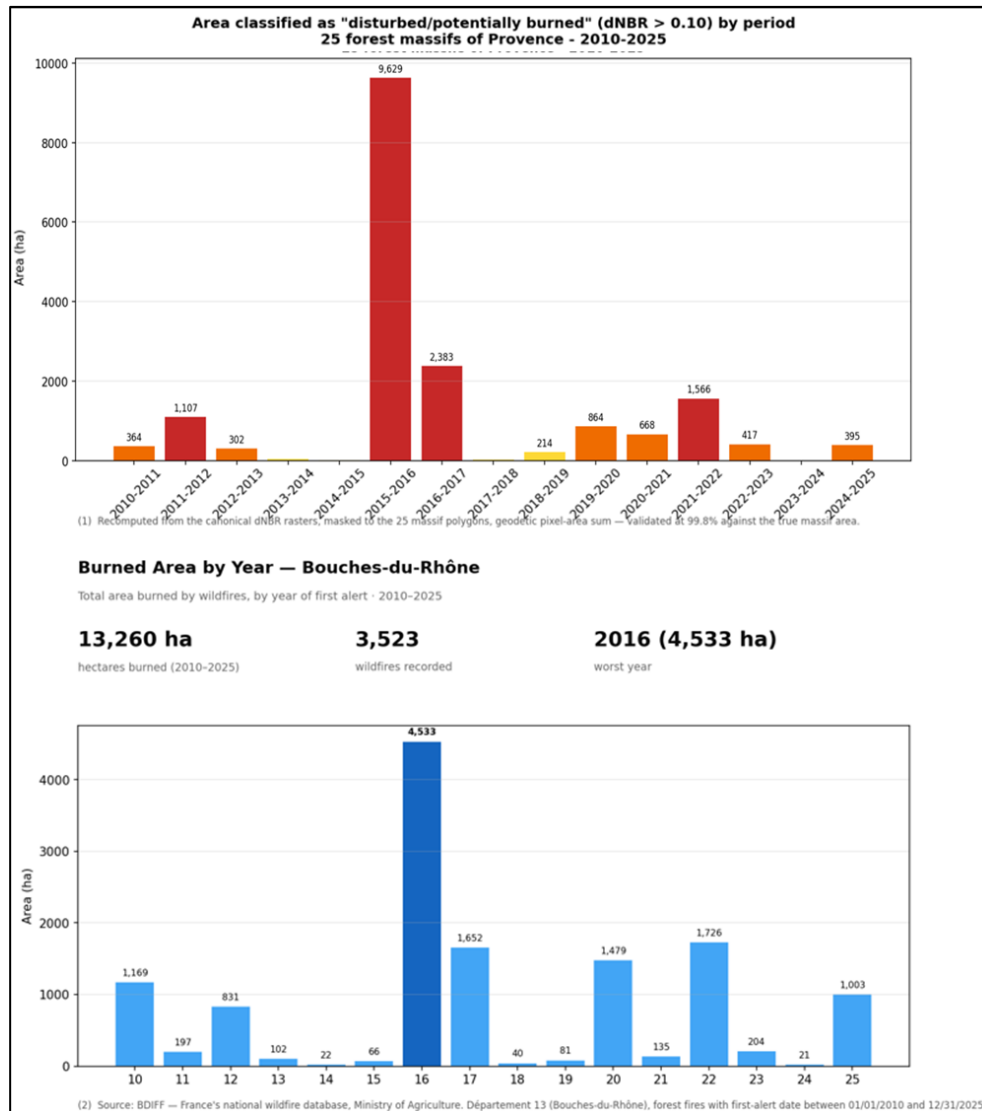


(a) Department-wide burn severity (dNBR2024-2025) across the 25 massifs. **(b)** Sentinel-1 VV/VH composite, May–June 2025 median — a single-period snapshot, not a pre/post-fire differential product.

Structural corroboration a marked drop in cross-polarized backscatter ($\Delta VH \approx -6$ dB) independently confirms, outside the optical domain, the immediate destruction of canopy volume during severe fire events.

05- RESULTS

Area classification (dNBR>0.1) vs Burned area BDIFF



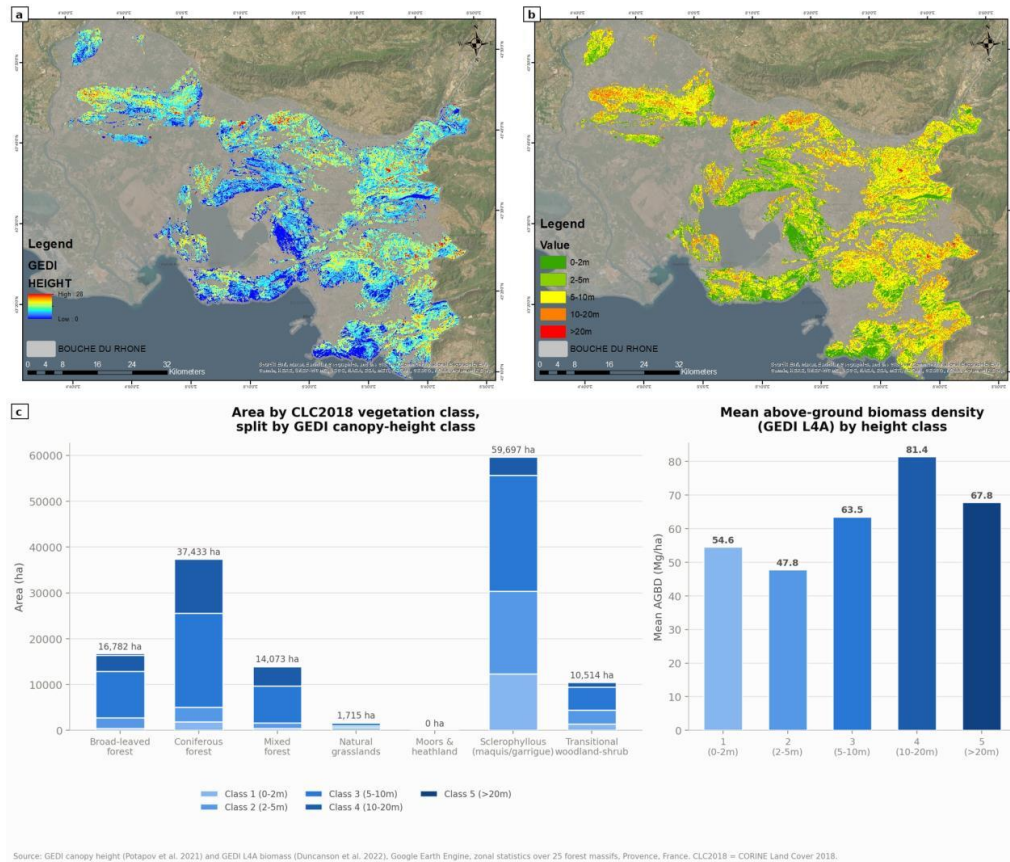
- dNBR > 0.10 'disturbed/potentially burned' area recomputed at annual resolution (vs. the 4-period grouping just shown), geodetic pixel-area sum validated at 99.8% against true massif area
- Cross-checked against BDIFF, France's national wildfire database: 13,260 ha burned across 3,523 recorded fires over 2010–2025, with 2016 the worst confirmed-fire year (4,533 ha)
- Year-to-year ranking broadly agrees between the two independent sources (2016, 2017 and 2022 stand out in both)

Reading the 2015-2016 gap The dNBR anomaly for 2015–2016 (9,629 ha) is roughly double the BDIFF-confirmed burned area for those same two years (4,599 ha) — consistent with, not contradicting, this period already being read as regionally synchronous drought browning rather than fire.

05- RESULTS

Vertical Structure: Canopy Height and Biomass (GEDI)

Figure 6. GEDI LiDAR Canopy Height and Biomass Structure



● Inland sub-humid massifs

Sainte-Baume, Concors

$$RH98 = 16.4 \pm 3.8 \text{ m} \cdot AGBD = 112.5 \text{ Mg} \cdot \text{ha}^{-1}$$

● Interior Mediterranean massifs

Sainte-Victoire, Alpilles

$$RH98 = 11.2 \pm 2.6 \text{ m} \cdot AGBD = 58.2 \text{ Mg} \cdot \text{ha}^{-1}$$

● Peri-urban / coastal massifs

Calanques, Étoile, Castillon

$$RH98 = 6.1 \pm 2.1 \text{ m} \cdot AGBD = 24.6 \text{ Mg} \cdot \text{ha}^{-1}$$

05- RESULTS

Forest Dynamics Index (IDF)

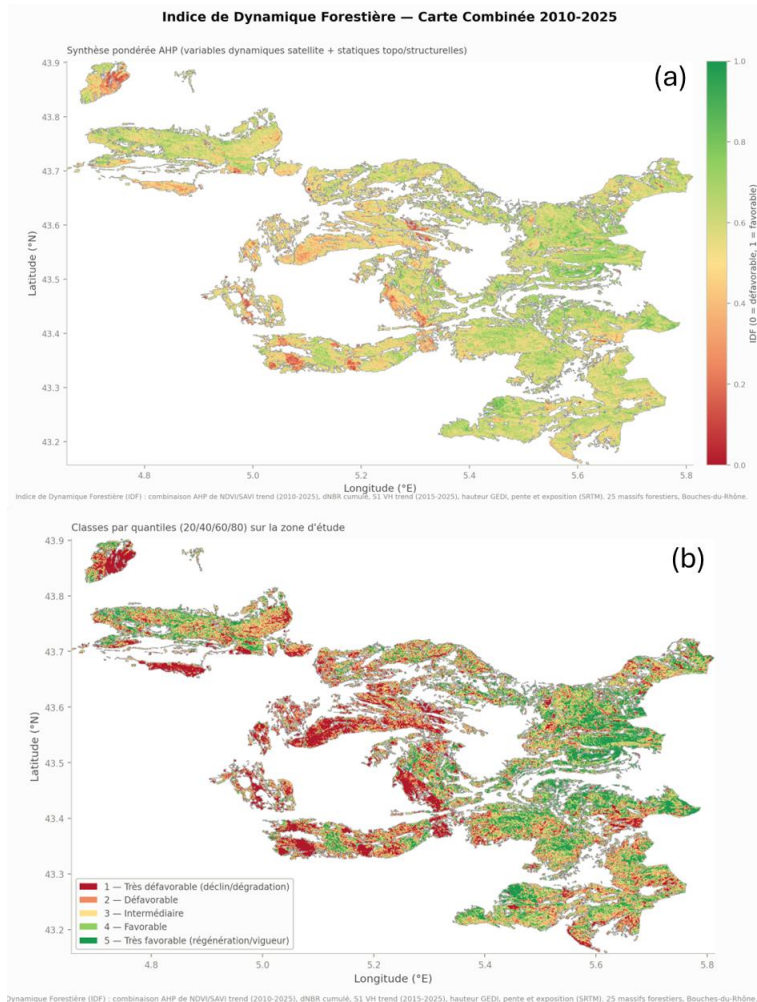
A weighted composite index (AHP)

- Combines: NDVI/SAVI trend (2010-2025), cumulative dNBR, Sentinel-1 VH trend (2014-2025), GEDI canopy height, slope and aspect
- 0 = unfavorable, 1 = favorable; panel (b) groups massifs into five discrete categories for management use
- Reproduces the resilience/vulnerability bifurcation seen in the trend and structural results: inland massifs score highest, dense-WUI coastal massifs score lowest

Formule :

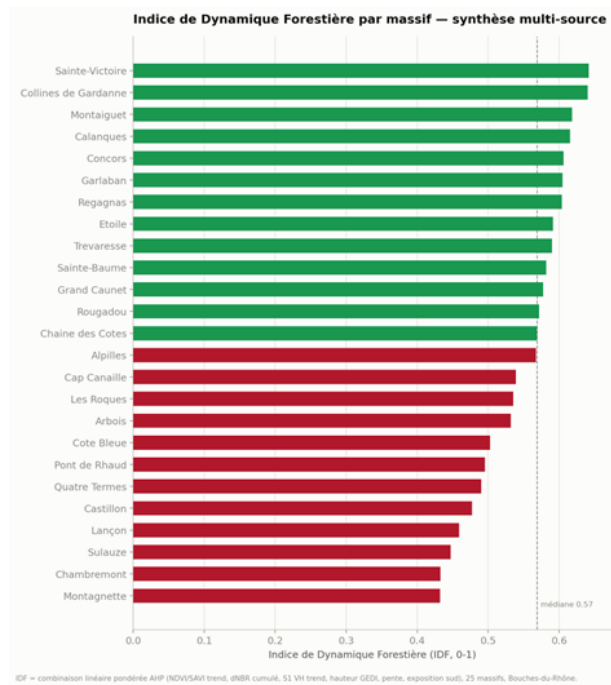
$$IDF = w1 \cdot NDVI_trend_norm + w2 \cdot SAVI_trend_norm + w3 \cdot (1 - dNBR_norm) + w4 \cdot VH_trend_norm + w5 \cdot GEDI_height_norm - w6 \cdot Slope_norm - w7 \cdot Southness_norm \text{ avec } \sum w_i = 1, \text{ poids issus de l'AHP}$$

Scientific precision The IDF carries no built-environment / housing-density term: it is an ecological dynamics index, not a WUI-specific risk layer.

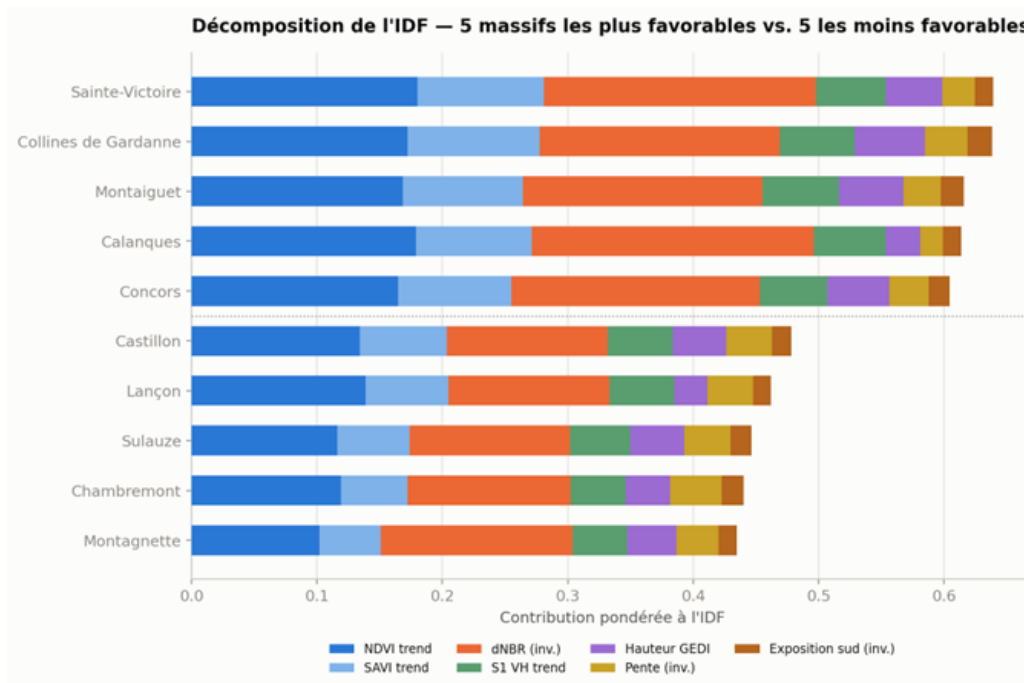


05- RESULTS

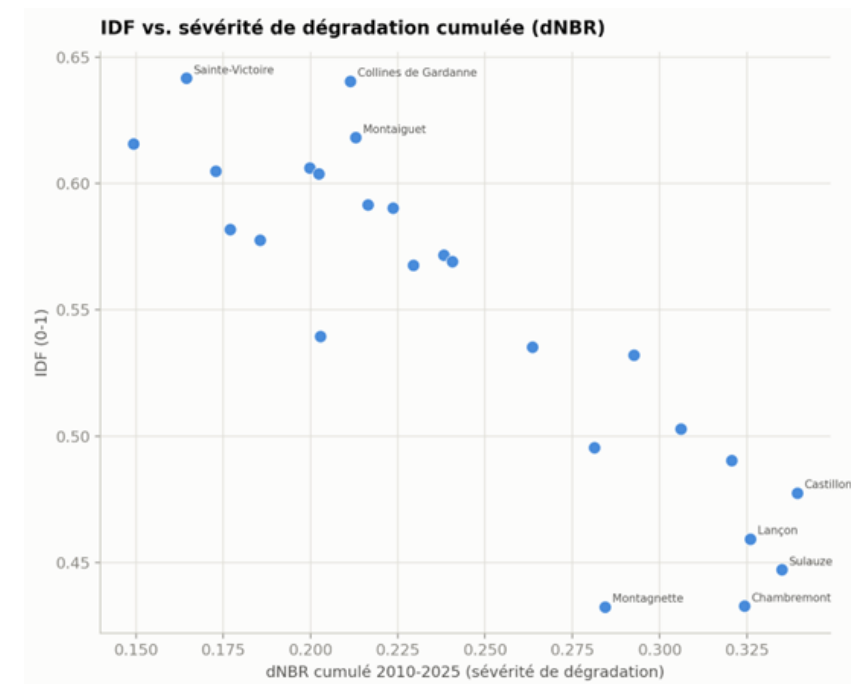
IDF Diagnostics: Ranking, Decomposition, Burn Severity



IDF ranking by massif (median 0.57)



Component decomposition, top 5 vs. bottom 5



IDF vs. cumulative dNBR

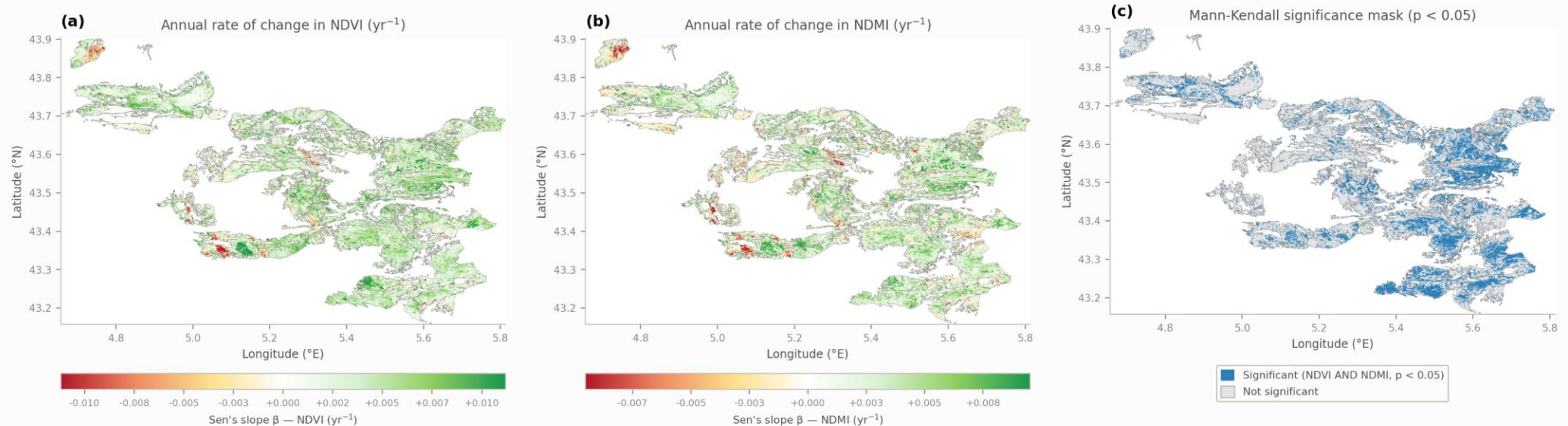
Open verification item In this IDF ranking, Calanques and Étoile score above the median — the opposite of their standing on raw β _NDVI (Table 5, slide 10). Flagged for cross-check against the underlying computation before the deposited version.

05- RESULTS

Spatially Explicit Trend Mapping: Sen's Slope and Man-Kendall

Department-wide NDVI and NDMI Sen's Slope Trend Maps (2010-2025)

Spatial distribution of Sen's slope (β) across the 25 forest massifs of Bouches-du-Rhône



Trend estimated via Theil-Sen robust slope on 16 annual Landsat composites (2010-2025, median NDVI/NDMI per year, L5/L7/L8/L9 SR, cloud/shadow-masked). Significance from a pairwise-sign Mann-Kendall test (S statistic, no tie correction); panel (c) shows pixels significant in both NDVI and NDMI. 25 forest massifs, Bouches-du-Rhône.

(a) NDVI Theil-Sen slope · (b) NDMI Theil-Sen slope · (c) Mann-Kendall significance mask ($p < 0.05$)

Beyond massif-aggregated statistics Pixel-level Theil-Sen and Mann-Kendall surfaces verify massif-level trends are not artefacts of spatial averaging. Significant pixels cluster in the western massif and the eastern/south-eastern part of the study area; several central massifs show sparser significance.

06- DISCUSSION

A Clear Spatial Ecological Bifurcation

Resilience — inland massifs

- Sainte-Baume, Sainte-Victoire, Concors: positive NDVI trend ($\beta > 0$) and localized greening
- Favorable topographic microclimates: deep karstic fissures retaining winter precipitation, rainfall > 800 mm
- Downy oak and post-fire Aleppo pine stands maintain stable biomass growth, corroborated by GEDI RH98 heights > 15 m

Chronic decline — coastal and peri-urban massifs

- Calanques, Étoile, Castillon, Marignane: severe chronic moisture deficit ($\beta_NDMI \leq -2.5 \times 10^{-3}/\text{yr}$) over more than 25% of surface
- Bioclimatic aridity: shallow limestone soils with negligible water storage, recurring summer heatwaves (2017, 2022)
- Anthropogenic recurrence: high fire frequency along the dense WUI, arresting ecological succession

06- DISCUSSION

Methodological Limitations and Sources of Uncertainty

- **Spatial resolution**

The 30 m Landsat resolution induces mixed-pixel effects along narrow fuel breaks, rocky outcrops, and fine-scale WUI housing clusters.

- **GEDI spatial gaps**

As a profiling LiDAR system, GEDI does not provide contiguous wall-to-wall coverage, requiring spatial interpolation for unsampled massif sections.

- **Landsat 7 SLC-off**

Pre-2013 Landsat 7 data required annual median compositing over multi-scene acquisitions to mitigate missing data lines.

06- DISCUSSION

Implications for Forest Management and DFCI Planning

The geostatistical trend maps and WUI risk classifications provide direct decision support for forestry and fire-prevention authorities.

1

Fuel Break Maintenance

Target rapid scrub encroachment and high fuel-load areas along strategic DFCI tracks.

2

Assisted Reforestation

Direct active restoration with drought-tolerant ecotypes toward zones exhibiting negative natural regeneration trajectories.

3

OLD Legal Enforcement

Guide the spatial enforcement of legal brush-clearing obligations across high-risk diffuse WUI zones.

06- CONCLUSION

Synthesis and Future Research Perspectives

Synthesis of Findings

- Quantified long-term canopy greenness (NDVI, SAVI) and moisture (NDMI) trajectories across 110,000 ha of Mediterranean forest, 2010-2025
- Decoupled abrupt wildfire destruction (e.g., Étoile 2016, Castillon 2020) from progressive bioclimatic drought dieback via Mann-Kendall / Theil-Sen and LandTrendr segmentation
- Identified peri-urban and coastal massifs (Calanques, Étoile, Castillon) as key vulnerability hotspots due to intense WUI pressure and shallow soil profiles
- Forest Dynamics Index (IDF)

Future Research Perspectives

Harmonized Landsat-Sentinel (HLS)

Fusing Sentinel-2 MSI data (10-20 m) from 2015 onward to refine spatial precision along wildland-urban interfaces.

Nationwide LiDAR HD Integration

Utilizing the ongoing IGN LiDAR HD campaign ($>10 \text{ pulses} \cdot \text{m}^{-2}$) to generate high-resolution 3D canopy models across all 25 massifs.

Predictive Deep Learning Modeling

Implementing spatiotemporal Convolutional-LSTM architectures to forecast forest canopy vulnerability under projected climate scenarios.

Thank You for Your Attention

Questions and Discussion

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